Development and Evaluation of Algorithmic Trading Strategies Using Mean-Reverting Stochastic Processes: Ornstein–Uhlenbeck and Cox–Ingersoll–Ross Models

MAC F244: Stochastic Calculus and Application to Finance (Semester I, AY 2026–2027)

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Abstract

Financial state variables governed by restorative economic forces—including cointegrated asset spreads, sovereign bond yields, and volatility indices—violate the unbounded variance growth of Geometric Brownian Motion. We formulate, calibrate, and empirically backtest continuous-time algorithmic trading strategies driven by the Ornstein–Uhlenbeck (OU) Gaussian diffusion and the Cox–Ingersoll–Ross (CIR) square-root diffusion. First, we derive closed-form transition densities, prove boundary non-attainment under the Feller condition ($2\kappa\theta > \xi^2$), and quantify Euler–Maruyama discretization biases against exact Gaussian and non-central chi-squared (χ'^2_d) sampling. Second, we implement exact analytical Maximum Likelihood Estimation (MLE) alongside AR(1) Ordinary Least Squares (OLS) for OU, and a Bessel-stabilized ($I^e_q(z) = e^{-z} I_q(z)$) two-stage MLE for CIR. Third, using 14 years of daily market data (N = 3,519, 2012–2025) with strict causal walk-forward evaluation (2015–2025), our trade-coherent OU pairs strategy on cointegrated `EWA/EWC` achieves a +25.88% net return (Sharpe 0.35) versus +6.20% (Sharpe 0.11) on non-cointegrated `BHP/RIO`, while our dual-CIR fixed-income regime strategy on `IEF/SHY` nearly doubles passive bond Sharpe (0.39 vs. 0.21) while cutting maximum drawdown from −23.92% to −16.71%. Combining both nearly uncorrelated strategies (r = 0.015) via convex optimization (`cvxpy`) yields a 0.49 net Sharpe ratio, 0.70 Sortino ratio, and −11.33% maximum drawdown.

I. Introduction & Literature

Classical equity derivatives theory models asset prices via Geometric Brownian Motion (GBM), $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, under which log-prices diffuse as non-stationary martingales with variance growing linearly in time, $\mathrm{Var}(\ln S_t) = \sigma^2 t$. However, empirical regularities across relative-value equity spreads, term-structure interest rates, and variance swaps exhibit pronounced *mean reversion*: macroeconomic substitution, arbitrage capital, and central-bank policy anchor trajectories around long-run equilibrium levels.

Continuous-time mean-reverting modeling originated with the Ornstein–Uhlenbeck (OU) process [1], adapted to short-rate term structures by Vasicek [2] and to statistical arbitrage pairs trading by Gatev et al. [6], Elliott et al. [5], and Leung & Li [10]. When two non-stationary $I(1)$ log-price processes $\ln P^{(A)}_t$ and $\ln P^{(B)}_t$ share a common stochastic trend in the sense of Engle & Granger [4], their linear combination $X_t = \ln P^{(A)}_t - \alpha - \beta \ln P^{(B)}_t$ is stationary $I(0)$, whose continuous-time scaling limit is the OU diffusion. Nevertheless, because OU innovations are state-independent, X_t admits negative values with positive probability—desirable for zero-mean spreads, yet flawed for strictly positive nominal yields or volatility indices. Cox, Ingersoll & Ross (CIR) [3] resolved this defect by introducing state-dependent diffusion $\xi\sqrt{r_t}\,dW_t$, inducing endogenous heteroskedasticity and strict positivity when $2\kappa\theta > \xi^2$.

This paper bridges stochastic calculus theory and institutional execution across four contributions: (i) rigorous discretization benchmarking of Euler schemes against exact $\mathcal{N}$ and χ'^2_d transitions near singular boundaries [8]; (ii) numerically overflow-free exact MLE calibration [7]; (iii) causal walk-forward trading rules eliminating rolling-intercept look-ahead illusions; and (iv) multi-asset empirical attribution across equity pairs (`EWA/EWC`, `GLD/GDX`, `BHP/RIO`) and Treasury ETFs (`IEF/SHY`).

II. Mathematical Formulation & Discretization

A. Ornstein–Uhlenbeck (OU) Diffusion

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ carry a standard Brownian motion W_t. The OU SDE for a cointegrated spread $X_t \in \mathbb{R}$ is

$$dX_t = \theta(\mu - X_t)\,dt + \sigma\,dW_t, \qquad \theta > 0,\ \mu \in \mathbb{R},\ \sigma > 0. \tag{1}$$

Applying Itô's Lemma to $Y_t = f(t, X_t) = e^{\theta t}(X_t - \mu)$, noting $\partial_{xx} f = 0$, gives $dY_t = \theta e^{\theta t}(X_t - \mu)\,dt + e^{\theta t} dX_t = \sigma e^{\theta t}\,dW_t$. Integrating over $[t, t+\Delta t]$ yields the strong solution:

$$X_{t+\Delta t} = \mu + (X_t - \mu)e^{-\theta\Delta t} + \sigma \int_t^{t+\Delta t} e^{-\theta(t+\Delta t - s)}\,dW_s. \tag{2}$$

Because the integrand is deterministic and square-integrable, the Wiener integral is Gaussian with zero conditional mean and variance given by Itô's Isometry:

$$X_{t+\Delta t} \mid X_t \sim \mathcal{N}\!\left(\mu + (X_t - \mu)e^{-\theta\Delta t},\ \frac{\sigma^2}{2\theta}\big(1 - e^{-2\theta\Delta t}\big)\right). \tag{3}$$

As $\Delta t \to \infty$, X_t converges in law to its invariant distribution $\mathcal{N}(\mu, \sigma^2_{\mathrm{eq}})$ where $\sigma_{\mathrm{eq}} = \sigma/\sqrt{2\theta}$, with expected mean-reversion half-life $t_{1/2} = \ln(2)/\theta$.

Euler–Maruyama Bias: First-order Euler discretization $X_{n+1} = X_n + \theta(\mu - X_n)\Delta t + \sigma\sqrt{\Delta t}\,Z_n$ ($Z_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$) coincides with the Milstein scheme since $\partial_x \sigma = 0$, attaining strong L^2 convergence order $O(\Delta t^{1.0})$ (empirically verified slope 1.06 in Fig. 1d). However, solving $V_{\mathrm{EM}} =$

$(1-\theta\Delta t)^2 V_{\text{EM}} + \sigma^2\Delta t$ reveals an upward stationary variance bias $V_{\text{EM}} = \frac{\sigma^2}{2\theta - \theta^2\Delta t} > \sigma^2_{\text{eq}}$ for $0 < \theta\Delta t < 2$, and numerical divergence for $\theta\Delta t \geq 2$. Exact sampling via (3) incurs zero discretization error at any Δt.

B. Cox–Ingersoll–Ross (CIR) Square-Root Process

For positive yield or volatility states $r_t \in \mathbb{R}_+$, the CIR SDE is

$$dr_t = \kappa(\theta - r_t)\,dt + \xi\sqrt{r_t}\,dW_t, \qquad \kappa, \theta, \xi > 0,\ r_0 > 0. \quad (4)$$

Proposition 1 (Feller Positivity Criterion)**.** *Let* $\tau_0 = \inf\{t \geq 0 : r_t = 0\}$. *Then* $\mathbb{P}(\tau_0 = \infty) = 1$ *iff* $2\kappa\theta \geq \xi^2$. *For* $0 < 2\kappa\theta < \xi^2$, *the origin is accessible* ($\mathbb{P}(\tau_0 < \infty) = 1$) *and instantaneously reflecting.*

Proof Sketch: Feller's scale density for (4) is $s(x) = \exp\left(-2\int_{x_0}^{x} \frac{\kappa(\theta-y)}{\xi^2 y}\,dy\right) \propto x^{-2\kappa\theta/\xi^2} e^{2\kappa x/\xi^2}$. The boundary 0 is unattainable iff $\int_0^{\varepsilon} s(x)\,dx = \infty$, which holds iff the exponent satisfies $2\kappa\theta/\xi^2 \geq 1$. When $\mathcal{F} \equiv 2\kappa\theta/\xi^2 < 1$, the speed density $m(x) = [\xi^2 x s(x)]^{-1} \propto x^{\mathcal{F}-1} e^{-2\kappa x/\xi^2}$ remains integrable near 0, yielding instantaneous reflection without absorption. □

Exact χ'^2 Law vs. Boundary Schemes: Conditional on r_t, $2c\,r_{t+\Delta t} \mid r_t \sim \chi'^2_d(\lambda_t)$ follows a non-central chi-squared distribution [3, 9] with

$$c = \frac{2\kappa}{\xi^2(1 - e^{-\kappa\Delta t})}, \quad d = \frac{4\kappa\theta}{\xi^2}, \quad \lambda_t = 2c\,r_t e^{-\kappa\Delta t}, \quad (5)$$

and stationary Gamma($\alpha = \frac{2\kappa\theta}{\xi^2}, \beta = \frac{2\kappa}{\xi^2}$) density, which is unimodal for $\mathcal{F} > 1$ but diverges at $r \to 0^+$ when $\mathcal{F} < 1$ (Fig. 1c). Because $\sqrt{r}$ violates Lipschitz continuity at $r = 0$, naive Euler steps turn negative whenever $Z_n < -\frac{r_n + \kappa(\theta - r_n)\Delta t}{\xi\sqrt{r_n \Delta t}}$. Comparing boundary remedies under stress ($\mathcal{F} = 0.34$, Fig. 1d), *Full Truncation* ($\tilde{r}_{n+1} = \tilde{r}_n + \kappa(\theta - \tilde{r}_n^+)\Delta t + \xi\sqrt{\tilde{r}_n^+}\Delta W_n$, $r_{n+1} = \tilde{r}_{n+1}^+$) [8] achieves 3.7× lower Wasserstein-1 (W_1) distributional error at $\Delta t = 1/4$ than Reflection ($|\cdot|$) and 1.6× lower than Absorption ($\max(0, \cdot)$).

III. Parameter Calibration Methodology

A. Analytical OU MLE vs. AR(1) OLS Regression

Given equispaced observations $\{X_0, \ldots, X_N\}$ at step Δt, (3) defines an exact Gaussian AR(1) transition $X_i = \alpha + \phi X_{i-1} + \varepsilon_i$, with $\phi = e^{-\theta\Delta t} \in (0,1)$, $\alpha = \mu(1-\phi)$, and $\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$ where $\sigma_\varepsilon^2 = \frac{\sigma^2}{2\theta}(1-\phi^2)$. Maximizing the conditional log-likelihood

$$\ell(\theta, \mu, \sigma) = -\frac{N}{2}\ln(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^{N}\left(X_i - \mu - (X_{i-1} - \mu)e^{-\theta\Delta t}\right)^2 \quad (6)$$

via $\nabla\ell = 0$ yields closed-form estimators in terms of sufficient sums $S_x = \sum X_{i-1}$, $S_y = \sum X_i$, $S_{xx} = \sum X_{i-1}^2$, $S_{xy} = \sum X_{i-1}X_i$:

$$\hat{\phi} = \frac{NS_{xy} - S_x S_y}{NS_{xx} - S_x^2}, \quad \hat{\mu} = \frac{S_y - \hat{\phi}S_x}{N(1-\hat{\phi})}, \quad \hat{\theta} = -\frac{\ln\hat{\phi}}{\Delta t}, \quad (7)$$

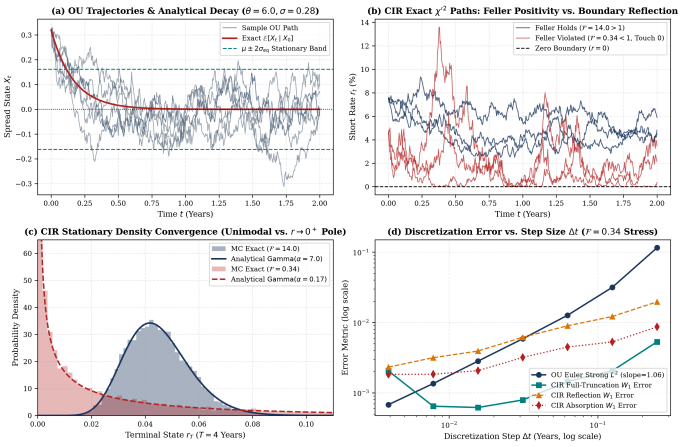


Figure 1: SDE simulation and discretization validation: (a) OU paths with exact mean decay; (b) CIR exact χ'^2 paths contrasting Feller-satisfied ($\mathcal{F} = 14.0$) vs. boundary-touching ($\mathcal{F} = 0.34$) regimes; (c) 12,000-path empirical vs. exact Gamma densities; (d) step-size convergence showing OU $O(\Delta t^{1.06})$ strong error and superior fidelity of CIR Full Truncation.

and $\hat{\sigma} = \hat{\sigma}_\varepsilon\sqrt{\frac{2\hat{\theta}}{1-\hat{\phi}^2}}$ with $\hat{\sigma}_\varepsilon^2 = \frac{1}{N}\sum_{i=1}^{N}(X_i - \hat{\alpha} - \hat{\phi}X_{i-1})^2$. By invariance of MLE under the bijection $(\mu, \phi) \leftrightarrow (\alpha, \phi)$, $(\hat{\theta}, \hat{\mu})$ coincide identically with exact-mapped AR(1) OLS (Table 1: $\hat{\theta} = 2.8778$ for both on `EWA/EWC`), differing only by $\sqrt{N/(N-2)}$ in $\hat{\sigma}$. Conversely, naive *Euler OLS* $\hat{\theta}_{\text{EM}} = (1-\hat{\phi})/\Delta t$ linearizes $e^{-\theta\Delta t} \approx 1 - \theta\Delta t$, systematically underestimating θ since $1 - e^{-x} < x$ for $x > 0$.

B. Bessel-Stabilized Exact CIR Maximum Likelihood

From (5), the CIR transition density for $r_i \mid r_{i-1}$ is

$$f(r_i \mid r_{i-1}) = c\,e^{-(u_i + v_i)}\left(\frac{v_i}{u_i}\right)^{q/2} I_q(2\sqrt{u_i v_i}), \quad (8)$$

where $u_i = c\,r_{i-1}e^{-\kappa\Delta t}$, $v_i = c\,r_i$, $q = \frac{2\kappa\theta}{\xi^2} - 1$, and $I_q(\cdot)$ is the modified Bessel function of the first kind. At daily frequency ($\Delta t = 1/252$), $c \sim O(10^4)$ drives $z_i = 2\sqrt{u_i v_i} > 10^3$, causing direct evaluation of $I_q(z_i) \sim e^{z_i}/\sqrt{2\pi z_i}$ to overflow IEEE-754 double precision. Following Kladívko [7], we introduce the exponentially scaled Bessel function $I_q^e(z) = e^{-z}I_q(z)$, transforming the log-likelihood algebraically into an overflow-free quadratic form:

$$\ell_{\text{CIR}} = \sum_{i=1}^{N}\Bigg[\ln c - \left(\sqrt{u_i} - \sqrt{v_i}\right)^2 + \frac{q}{2}\left(\ln\frac{r_i}{r_{i-1}} + \kappa\Delta t\right) + \ln I_q^e(z_i)\Bigg]. \quad (9)$$

We initialize $(\kappa_0, \theta_0, \xi_0)$ via Weighted Conditional Least Squares (dividing Δr_i by $\sqrt{r_{i-1}}$ to remove heteroskedasticity) before maximizing (9) in log-parameter space via L-BFGS-B. Figure 2 confirms $O(N^{-1/2})$ Monte Carlo RMSE decay, Gaussianity of `EWA/EWC` OU innovations, and empirical $\xi\sqrt{r_t}$ scaling.

IV. Trading Strategy & Empirical Results

A. Dataset, Cointegration Screening & Causal Protocol

We ingest $N = 3{,}519$ daily adjusted observations from Yahoo Finance (`yfinance`) spanning January 3, 2012 to December 30, 2025 across commodity-currency equity ETFs

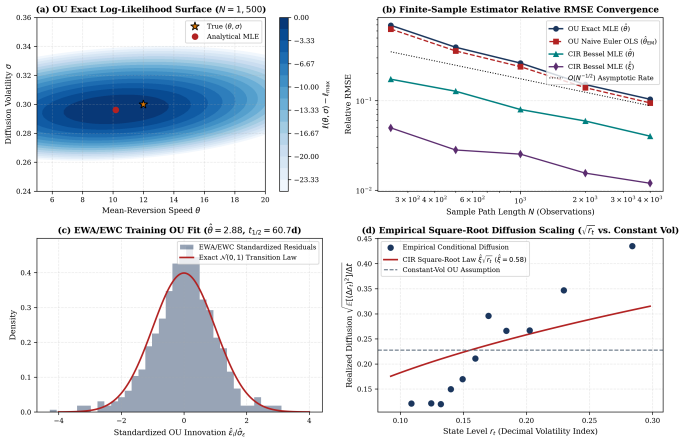


Figure 2: Calibration diagnostics: (a) global concavity of OU log-likelihood; (b) $O(N^{-1/2})$ relative RMSE consistency across 200 MC trials; (c) `EWA/EWC` standardized residual normality; (d) empirical verification of CIR $\sqrt{r_t}$ volatility scaling against constant-volatility OU.

Table 1: In-Sample Calibration Estimates (2012–2014 Training Split, $N = 756$)

Series / Model	Method	$\hat{\theta}$ / $\hat{\kappa}$	$\hat{\mu}$ / $\hat{\theta}_{\text{CIR}}$	$\hat{\sigma}$ / $\hat{\xi}$	$t_{1/2}$ (d)	$\ell_{\max}$ / $\mathcal{F}$
`EWA/EWC` (OU)	Exact MLE	2.878	0.0058	0.1215	60.7	2611.5
`EWA/EWC` (OU)	AR(1) OLS	2.878†	0.0058	0.1217	60.7	$R^2 = 0.98$
10Y Yield `^TNX`	CIR MLE	1.363	0.0224	0.0479	128.2	$\mathcal{F} = 26.6$
`^TNX` Rate-Vol	CIR MLE	7.816	0.0068	0.0728	22.3	$\mathcal{F} = 20.1$
CBOE `^VIX`	CIR MLE	21.576	0.1535	0.4467	8.1	$\mathcal{F} = 33.2$

†Exact OLS map matches MLE; naive Euler OLS yields $\hat{\theta}_{\text{EM}} = 2.861$ (downward bias).

(`EWA`, `EWC`), precious-metal/mining ETFs (`GLD`, `GDX`), global miners (`BHP`, `RIO`), Treasury bond ETFs (`IEF` 7–10Y, `SHY` 1–3Y), 10-Year US Treasury yields (`^TNX`), and `^VIX`. Isolated bond-market holiday gaps (1 observation in `^TNX`) are causally forward-filled. Days 1–756 (2012–2014) serve exclusively as training warmup; all backtests run strictly out-of-sample over January 6, 2015 to December 30, 2025 ($N_{\text{OOS}} = 2{,}763$ trading days, ≈ 11 years), executing day-t close signals at $t + 1$ (`execution_lag = 1`).

Full-sample Engle–Granger cointegration tests confirm statistically significant cointegration for `EWA/EWC` ($t = -3.53, p = 0.0299$, ADF $p = 0.0073$, $R^2 = 0.948$), marginal cointegration for `GLD/GDX` ($t = -2.96, p = 0.121$), and non-cointegration for `BHP/RIO` ($t = -2.28, p = 0.386$).

B. Strategy Architecture & Convex Multi-Asset Sizing

1) Trade-Coherent OU Pairs Strategy: On a causal $W = 180$-day rolling window, we estimate OLS log-hedge parameters $(\hat{\alpha}_t, \hat{\beta}_t)$ and analytical OU MLE $(\hat{\theta}_t, \hat{\mu}_t, \hat{\sigma}_t)$, forming the standardized score $Z_t = (X_t - \hat{\mu}_t)/\hat{\sigma}_{\text{eq},t}$. A critical implementation safeguard is *trade-coherent hedge locking*: updating rolling $\hat{\alpha}_t$ inside an open trade absorbs price divergence into the intercept, creating illusory mean-reversion exits at a loss. Locking $(\hat{\alpha}_{t_0}, \hat{\beta}_{t_0}, \hat{\mu}_{t_0}, \hat{\sigma}_{\text{eq},t_0})$ from trade entry t_0 until exit ensures Z_t measures true mark-to-market convergence while eliminating daily beta-rebalancing churn. Hysteresis rules enter Long $(+1)$ at $Z_t \le -1.50$ and Short (-1) at $Z_t \ge +1.50$ (gated by stationarity filter $t_{1/2} \le 65$d), exiting at $|Z_t| \le 0.25$ or stop-loss $|Z_t| \ge 3.25$ with post-stop cooldown, normalized to $1.5\times$ gross exposure net of 5 bps one-way slippage and 25 bps/yr short-borrow fee.

2) Dual-CIR Fixed-Income Regime Strategy: Because nominal yields undergo multi-year monetary cycles

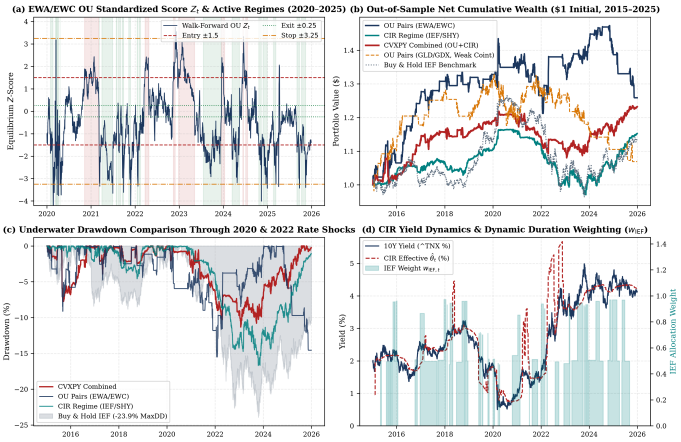


Figure 3: Out-of-sample empirical performance (2015–2025): (a) `EWA/EWC` OU Z_t triggers and Long/Short episodes; (b) net cumulative wealth trajectories; (c) underwater drawdowns through the 2020 COVID shock and 2022 Fed tightening cycle; (d) CIR yield vs. effective $\hat{\theta}_t$ and defensive `IEF` de-risking across 2021–2022.

Table 2: Out-of-Sample Net Performance Attribution (2015–2025, $N = 2{,}763$)

Strategy	Ret%	CAGR%	Vol%	Sharpe	Sortino	MaxDD%	PF	Drag%
CVXPY (OU+CIR)	**23.20**	**1.92**	**4.07**	**0.49**	**0.70**	**-11.33**	**1.12**	**0.53**
CIR Regime (`IEF/SHY`)	15.26	1.30	3.47	0.39	0.58	-16.71	2.50	4.73
OU Pairs (`EWA/EWC`)	25.88	2.12	6.69	0.35	0.53	-15.50	1.47	11.06
OU Pairs (`GLD/GDX`)	6.91	0.61	6.47	0.13	0.18	-19.51	1.23	7.25
OU Pairs (`BHP/RIO`)	6.20	0.55	8.01	0.11	0.16	-18.68	1.21	6.99
Buy & Hold `IEF`	13.49	1.16	6.63	0.21	0.30	-23.92	1.04	0.00

($t_{1/2} = 128$d) whereas 21-day realized yield volatility v_t mean-reverts rapidly ($t_{1/2} = 22.3$d, ADF $p = 0.0047$), we calibrate dual CIR models on $r_t = $ `^TNX`$_t/100$ and v_t ($W = 252$d). Using heteroskedastic CIR scaling $\sigma_{\text{CIR}}(r_t) = \sqrt{(\hat{\xi}_r^2/2\hat{\kappa}_r)r_t}$, the strategy overweights `IEF` with inverse-root volatility sizing $w_{\text{IEF}} = \text{clip}(\sqrt{\hat{\theta}_t/r_t}, 0.70, 1.15)$ when yields are attractively elevated ($1.0 \le Z_t^{\text{CIR}} < 2.25$) and rate volatility is calm ($v_t \le 1.15\hat{\theta}_{v,t}$), rotates 100% into short-duration `SHY` carry during low-yield bubbles ($Z_t^{\text{CIR}} \le -0.85$) or rate-volatility breakouts ($Z_t^{\text{CIR}} \ge 2.25$ or $v_t > 1.15\hat{\theta}_{v,t}$), and holds a 50% vol-scaled core otherwise (3 bps cost).

3) `cvxpy` Portfolio Allocation: Every 21 trading days, we solve the causal convex program $\min_w \frac{\gamma}{2} w^\top \hat{\Sigma}_t w - \hat{\mu}_t^\top w + \lambda_{\text{tc}} \|w - w_{t-1}\|_1$ s.t. $\mathbf{1}^\top w = 1, w \in [0.15, 0.85]^2$ ($\gamma = 8, \lambda_{\text{tc}} = 2\times10^{-3}$) combining `OU Pairs (EWA/EWC)` and `CIR Regime (IEF/SHY)`.

C. Empirical Interpretation & Sensitivity Analysis

Table 2 and Figures 3–4 reveal four central findings:

- **Cointegration Rank Monotonicity:** Across identically parameterized OU pairs strategies, net return, Sharpe ratio, and trade velocity align monotonically with Engle–Granger cointegration strength: strongly cointegrated `EWA/EWC` ($p = 0.030$) delivers +25.88% net return (Sharpe 0.35, 54.9% win rate, 20.8-day average hold), outperforming weakly cointegrated `GLD/GDX` (+6.91%, Sharpe 0.13, 33.9d hold) and non-cointegrated `BHP/RIO` (+6.20%, Sharpe 0.11, 41.7d hold).
- **CIR Rate-Shock Protection:** During the historic 2021–2023 bond bear market, passive `IEF` suffered a −23.92% drawdown. The dual-CIR strategy detected

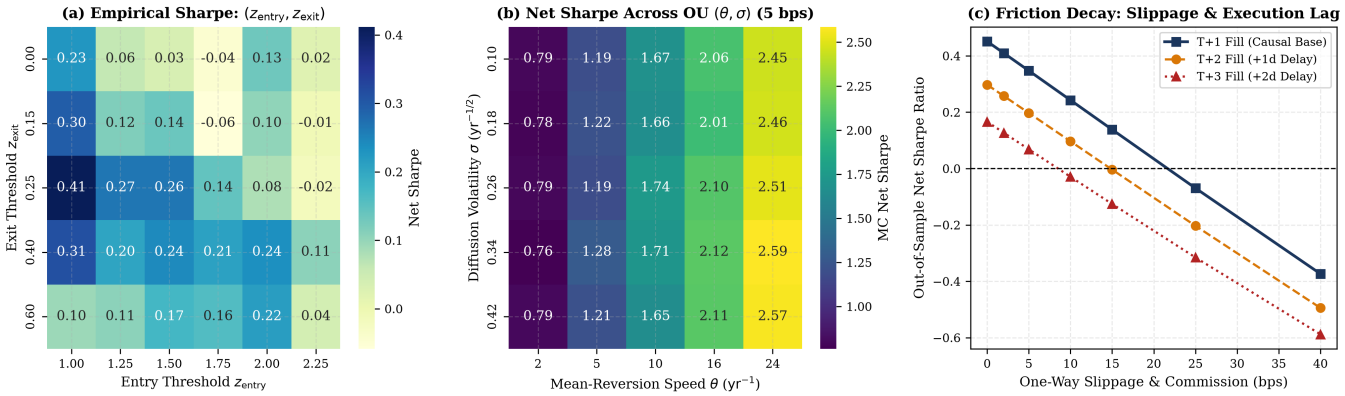


Figure 4: Multi-dimensional sensitivity analysis: (a) empirical out-of-sample Sharpe ratio across $(z_{\text{entry}}, z_{\text{exit}})$ on `EWA/EWC`; (b) Monte Carlo net Sharpe across OU (θ, σ) parameter space at 5 bps cost; (c) net Sharpe decay across one-way transaction costs (0–40 bps) and execution lags $(T+1, T+2, T+3)$.

both sub-equilibrium 2020–2021 yields and 2022 rate-volatility stress (Fig. 3d), shifting into `SHY` to cap maximum drawdown at -16.71% while nearly doubling annualized Sharpe (0.39 vs. 0.21, profit factor 2.50).

- **Cross-Asset Diversification:** Daily returns of `OU Pairs (EWA/EWC)` and `CIR Regime (IEF/SHY)` exhibit $r = 0.0147$ correlation. Consequently, the `cvxpy` allocator elevates net Sharpe to 0.49 (Sortino 0.70) and compresses maximum drawdown to -11.33%.
- **Parameter & Friction Trade-Offs (Fig. 4):** Synthetic surfaces over (θ, σ) confirm that net Sharpe scales strongly with $\sqrt{\theta}$ (rising from 0.79 at $\theta = 2$ to 2.59 at $\theta = 24$) while remaining nearly invariant to σ once thresholds scale with σ_{eq}. Empirically, `EWA/EWC` remains profitable up to ≈ 20 bps one-way cost (40 bps round-trip per leg, break-even near 21 bps at $T+1$), while adding +1d and +2d execution delays lowers 5-bps Sharpe from 0.35 to 0.20 and 0.07, underscoring timely close execution.

V. Conclusion & Risk Limitations

Modeling mean-reverting financial spreads and sovereign yields via continuous-time OU and CIR diffusions provides principled entry normalization (σ_{eq} and $\sigma_{\text{CIR}}(r_t)$), exact likelihood calibration, and disciplined regime risk control. Nonetheless, live deployment faces four structural limitations: (i) *cointegration breakdown and hedging error*, where macroeconomic shifts (e.g., diverging commodity exports between Australia and Canada in late 2024–2025, visible in Fig. 3b) permanently alter β_t, necessitating Kalman-filtered dynamic hedges and strict stop-losses; (ii) *monetary regime shifts*, where deterministic central-bank tightening violates time-homogeneous CIR drift $\kappa(\theta - r_t)$, motivating jump-diffusion or Markov-switching extensions; (iii) *fat-tailed innovations*, where earnings or macro announcements induce discontinuous jumps exceeding Gaussian/χ'^2 tails; and (iv) *short-sale frictions*, where stock-loan fee spikes or recalls during market stress asymmetrically penalize long-horizon convergence trades.

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